

ElevanceSkills AI Engineering Suite - Documentation

1. Problem Statement

Deploying standalone conversational interfaces presents integration fragmentations, redundant token allocations, and lack of cross-module context. This unified engineering suite addresses these issues by binding individual large language model architectures into a single workspace panel. It covers multi-modal diagnostics, RAG knowledge ingestion, localized translation mapping, research querying, and sentiment support tracking.

2. Unified Portfolio Architecture

Module Directory	Functional Scope / Task Feature	Primary Stack Elements
gemini_chatbot/	RAG engine parsing Power BI context patterns and factual extraction strings.	FAISS Index, LangChain Vector Stores, Gemini SDK
multi-modal/	Vision reasoning processing complex image data alongside textual tokens.	Gemini Flash Multi-Modal Vision API
med_qa_chatbot/	Clinical QA application featuring direct authentication mappings.	Dotenv, Google GenerativeAI API

Module Directory	Functional Scope / Task Feature	Primary Stack Elements
arxiv_expert_chatbot/	Kaggle Research Paper Parsing and visual discovery platform.	arXiv Query Formats, DataFrames
multilingual_chatbot/	Cross-lingual localization layers and real-time localized synthesis.	Streamlit UI Layouts, Language Mappings
sentiment_chatbot/	Sentiment-aware tracking platform for routing support prioritization logs.	Text Blob / Analytical Classifier Scoring

3. Preprocessing, Feature Engineering, and Model Selection

- **Knowledge Indexing Optimization:** Raw platform data was converted using semantic text splitters to prevent vector buffer overruns during indexing.
- Factual Validation Assertions:** Token extraction modules were updated with robust defensive string checking to block runtime null reference errors.
- Explicit API Configuration:** Decoupled implicit Application Default Credentials (ADC) token structures in favor of targeted local environment variable loaders to eliminate 401 unauthenticated request faults.

4. Execution & Setup Reference

To initialize the parent unified workspace matrix dashboard, run the following command from the workspace root directory:

```
python -m streamlit run app.py
```